

Assignment #1 | LP Problems Solver

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1. Simplex Method:

Code Explanation:

simplex:

- Adds slack variables.
- Calls `handleUrVariables()` to handle unrestricted variables.
- Forms Tableau.

handleUrVariables:

- Replaces x_i with $x_i^+ - x_i^-$

solveTableau:

- Implements the Simplex algorithm

Sample Run:

Linear Programming

Goal Programming

Enter number of Decision Variables

4

Enter number of Constraints

3

Generate Model

Optimization direction

Minimize

Objective function $Z =$

5

-4

6

8

Please fill in your constraints

1

2

2

4

$\leq$

40

2

-1

1

2

$\leq$

8

4

-2

1

-1

$\leq$

10

Unrestricted Variables

☐ x1

☐ x2

☐ x3

☐ x4

Solve

Step 1

x1	x2	x3	x4	s1	s2	s3	RHS
5	-4	6	8	0	0	0	0
1	2	2	4	1	0	0	40
2	-1	1	2	0	1	0	8
4	-2	1	-1	0	0	1	10

Unrestricted Variables

☐ x1

☐ x2

☐ x3

☐ x4

Solve

Step 1

x1	x2	x3	x4	s1	s2	s3	RHS
5	-4	6	8	0	0	0	0
1	2	2	4	1	0	0	40
2	-1	1	2	0	1	0	8
4	-2	1	-1	0	0	1	10

Step 2

x1	x2	x3	x4	s1	s2	s3	RHS
7	0	10	16	2	0	0	80
0.5	1	1	2	0.5	0	0	20
2.5	0	2	4	0.5	1	0	28
5	0	3	3	1	0	1	50

The Solution is feasible

objective = 80

x2 = 20

s2 = 28

s3 = 50

2. Big M Method:

Code Explanation:

formulateConstraints:

- Handles unrestricted variables.
- Adds slack, surplus and artificial variables.

simplex:

- Forms Tableau.
- Implements the Simplex algorithm.

Sample Run:

Linear Programming

Goal Programming

Enter number of Decision Variables

Enter number of Constraints

Generate Model

Optimization direction

Minimize

Objective function $Z =$

Please fill in your constraints

1

1

1

=

7

2

-5

1

≥

10

Unrestricted Variables ☐ X1 ☐ X2 ☐ X3

Big M Method

Solve

Step 1

x1	x2	x3	a1	e2	a2	RHS
-3000001	3999998	-2000001	0	1000000	0	-17000000
1	1	1	1	0	0	7
2	-5	1	0	-1	1	10

Step 2

Step 1

x1	x2	x3	a1	e2	a2	RHS
-3000001	3999998	-2000001	0	1000000	0	-17000000
1	1	1	1	0	0	7
2	-5	1	0	-1	1	10

Step 2

x1	x2	x3	a1	e2	a2	RHS
0	-3500004.5	-500000.5	0	-500000.5	1500000.5	-1999995
0	3.5	0.5	1	0.5	-0.5	2
1	-2.5	0.5	0	-0.5	0.5	5

Step 3

x1	x2	x3	a1	e2	a2	RHS
0	0	0.1428571428405121	1000001.2857142857	0.1428571428405121	999999.8571428572	7.571428571362048
0	1	0.14285714285714285	0.2857142857142857	0.14285714285714285	-0.14285714285714285	0.5714285714285714
1	0	0.8571428571428571	0.7142857142857142	-0.1428571428571429	0.1428571428571429	6.428571428571429

The Solution is feasible

objective = 7.571428571362048
x2 = 0.5714285714285714
x1 = 6.428571428571429

3. Two Phase Method:

Code Explanation:

formulateTwoPhase:

- Adds slack, surplus and artificial variables.
- Handles unrestricted variables.

TwoPhasesimplex:

- Implements Phase 1.

Secondarysimplex:

- Implements Phase 2.
- Converts problem into standard simplex tableau.

Sample Run:

Linear Programming

Goal Programming

Enter number of Decision Variables

Enter number of Constraints

Generate Model

Optimization direction

Minimize

Objective function Z =

Please fill in your constraints

1

1

1

=

7

2

-5

1

≥

10

Unrestricted Variables ☐ x1 ☐ x2 ☐ x3

Two Phase Method

Solve

Step 1

x1	x2	x3	a1	e2	a2	RHS
-3	4	-2	0	1	0	-17
1	1	1	1	0	0	7
2	-5	1	0	-1	1	10

Step 2

Linear Programming

Goal Programming

Step 2

x1	x2	x3	a1	e2	a2	RHS
0	-3.5	-0.5	0	-0.5	1.5	-2
0	3.5	0.5	1	0.5	-0.5	2
1	-2.5	0.5	0	-0.5	0.5	5

Step 3

x1	x2	x3	a1	e2	a2	RHS
0	0	0	1	0	1	0
0	1	0.14285714285714285	0.2857142857142857	0.14285714285714285	-0.14285714285714285	0.5714285714285714
1	0	0.8571428571428571	0.7142857142857142	-0.1428571428571429	0.1428571428571429	6.428571428571429

Step 4

x1	x2	x3	e2	RHS
-1	-2	-1	0	0
0	1	0.14285714285714285	0.14285714285714285	0.5714285714285714
1	0	0.8571428571428571	-0.1428571428571429	6.428571428571429

Step 5

x1	x2	x3	e2	RHS
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Linear Programming

Goal Programming

x1	x2	x3	a1	e2	a2	RHS
0	0	0	1	0	1	0
0	1	0.14285714285714285	0.2857142857142857	0.14285714285714285	-0.14285714285714285	0.5714285714285714
1	0	0.8571428571428571	0.7142857142857142	-0.1428571428571429	0.1428571428571429	6.428571428571429

Step 4

x1	x2	x3	e2	RHS
-1	-2	-1	0	0
0	1	0.14285714285714285	0.14285714285714285	0.5714285714285714
1	0	0.8571428571428571	-0.1428571428571429	6.428571428571429

Step 5

x1	x2	x3	e2	RHS
0	0	0.1428571428571428	0.1428571428571428	7.571428571428571
0	1	0.14285714285714285	0.14285714285714285	0.5714285714285714
1	0	0.8571428571428571	-0.1428571428571429	6.428571428571429

The Solution is feasible

objective = 7.571428571428571
x2 = 0.5714285714285714
x1 = 6.428571428571429

4. Goal Programming:

Code Explanation:

`solve_goal_programming:`

- Create constraint matrix.
- Set objective function.
- Call simplex method to solve tableau.

Sample Run:

Linear Programming

Goal Programming

Enter number of Decision Variables:

Enter number of Constraints:

Enter number of Goals:

Generate Model

Phase 1 fill in your constraints

1500

3000

≤

15000

200

0

≤

1000

100

400

≥

1200

0

250

≤

800

Phase 2 fill in your goals with the higher priority ones on top

200

0

≤

1000

100

400

≥

1200

0

250

≤

800

Unrestricted Variables: ☐ x1 ☐ x2

Solve

Step 1

x1	x2	e1	s1	e2	s2	e3	s3	e4	s4	RHS
0	0	-1	0	0	0	0	0	0	0	0
200	0	0	0	-1	0	0	0	0	0	1000
100	400	0	0	0	0	-1	0	0	0	1200
0	250	0	0	0	0	0	0	-1	0	800
1500	3000	-1	1	0	0	0	0	0	0	15000
200	0	0	0	-1	1	0	0	0	0	1000
100	400	0	0	0	0	-1	1	0	0	1200
0	250	0	0	0	0	0	0	-1	1	800

Step 2

x1	x2	e1	s1	e2	s2	e3	s3	e4	s4	RHS
0	0	-1	0	0	0	0	0	0	0	0
200	0	0	0	-1	0	0	0	0	0	1000
100	400	0	0	0	0	-1	0	0	0	1200

Linear Programming	
Goal Programming	

Step 2

x1	x2	e1	s1	e2	s2	e3	s3	e4	s4	RHS
0	0	-1	0	0	0	0	0	0	0	0
200	0	0	0	-1	0	0	0	0	0	1000
100	400	0	0	0	0	-1	0	0	0	1200
0	250	0	0	0	0	0	0	-1	0	800
1500	3000	-1	1	0	0	0	0	0	0	15000
200	0	0	0	-1	1	0	0	0	0	1000
100	400	0	0	0	0	-1	1	0	0	1200
0	250	0	0	0	0	0	0	-1	1	800

Step 3

x1	x2	e1	s1	e2	s2	e3	s3	e4	s4	RHS
0	0	-1	0	0	0	0	0	0	0	0
0	0	0	0	0	-1	0	0	0	0	0
0	400	0	0	0.5	-0.5	-1	0	0	0	700
0	250	0	0	0	0	0	0	-1	0	800
0	3000	-1	1	7.5	-7.5	0	0	0	0	7500
1	0	0	0	-0.005	0.005	0	0	0	0	5
0	400	0	0	0.5	-0.5	-1	1	0	0	700
0	250	0	0	0	0	0	0	-1	1	800

Step 4

x1	x2	e1	s1	e2	s2	e3	s3	e4	s4	RHS
0	0	-1	0	0	0	0	0	0	0	0
0	0	0	0	0	-1	0	0	0	0	0
0	0	0	0	0	0	0	-1	0	0	0
0	0	0	0	-0.3125	0.3125	0.625	-0.625	-1	0	362.5
0	0	-1	1	3.75	-3.75	7.5	-7.5	0	0	2250

Linear Programming
Goal Programming

0	3000	-1	1	7.5	-7.5	0	0	0	7500
1	0	0	0	-0.005	0.005	0	0	0	5
0	400	0	0	0.5	-0.5	-1	1	0	700
0	250	0	0	0	0	0	0	-1	1
									800

Step 4

x1	x2	e1	s1	e2	s2	e3	s3	e4	s4	RHS
0	0	-1	0	0	0	0	0	0	0	0
0	0	0	0	0	-1	0	0	0	0	0
0	0	0	0	0	0	0	-1	0	0	0
0	0	0	0	-0.3125	0.3125	0.625	-0.625	-1	0	362.5
0	0	-1	1	3.75	-3.75	7.5	-7.5	0	0	2250
1	0	0	0	-0.005	0.005	0	0	0	0	5
0	1	0	0	0.00125	-0.00125	-0.0025	0.0025	0	0	1.75
0	0	0	0	-0.3125	0.3125	0.625	-0.625	-1	1	362.5

Step 5

x1	x2	e1	s1	e2	s2	e3	s3	e4	s4	RHS
0	0	-1	0	0	0	0	0	0	0	0
0	0	0	0	0	0	-1	0	0	0	0
0	0	0	0	0	0	0	0	-1	0	0
0	0	0.08333333333333333	-0.08333333333333333	-0.625	0.625	0	0	-1	0	175
0	0	-0.13333333333333333	0.13333333333333333	0.5	-0.5	1	-1	0	0	300
1	0	0	0	-0.005	0.005	0	0	0	0	5
0	1	-0.0003333333333333333	0.0003333333333333333	0.0025	-0.0025	0	0	0	0	2.5
0	0	0.08333333333333333	-0.08333333333333333	-0.625	0.625	0	0	-1	1	175

The solution is feasible!

e3 = 300
x1 = 5
x2 = 2.5
s4 = 175

Loss of goal 1 = 0 / No Loss
Loss of goal 2 = 0 / No Loss
Loss of goal 3 = 175 / Loss Here